

## Summative Assignment

|                                 |                                                       |
|---------------------------------|-------------------------------------------------------|
| <b>Module code and title</b>    | COMP4097 Advanced Computer Graphics and Visualisation |
| <b>Academic year</b>            | 2025-26                                               |
| <b>Coursework title</b>         | AdvComputer Graphics                                  |
| <b>Coursework credits</b>       | 5 credits                                             |
| <b>% of module's final mark</b> | 50%                                                   |
| <b>Lecturer</b>                 | Frederick Li                                          |
| <b>Submission date*</b>         | Thursday, 19 February, 2026 14:00                     |
| <b>Estimated hours of work</b>  | 10                                                    |
| <b>Submission method</b>        | Ultra                                                 |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Additional coursework files</b>           | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Required submission items and formats</b> | <i>You should compress all files into a single zip file and upload it to Blackboard Ultra for submission. Your submission should include the two-page A4 report, your implementation with all source code and resource files, a readme file with instructions on how to run your game and details of any external resources you have used (if applicable), and the required 1-minute video. (Note: Ensure that you include all necessary source code and resource files in your submission, enabling me to run your implementation on my machine. Failure to provide these materials may result in a zero mark.)</i> |

\* This is the deadline for all submissions except where an approved extension is in place. For assessments carried out in weekly practicals or other scheduled sessions, the date shown will be the Monday of the week in which the assessments take place.

Late submissions received within 5 working days of the deadline will be capped at 40%.

Late submissions received later than 5 days after the deadline will receive a mark of 0.

It is your responsibility to check that your submission has uploaded successfully and obtain a submission receipt.

Your work must be done by yourself (or your group, if there is an assigned groupwork component) and comply with the university rules about plagiarism and collusion. Students suspected of plagiarism, either of published or unpublished sources, including the work of other students, or of collusion will be dealt with according to University guidelines (<https://www.dur.ac.uk/learningandteaching.handbook/6/2/4/>).

## Module: COMP4097 Advanced Computer Graphics

### Summative Assignment 2025-26

**Title:** *The Durham Lumiere Simulation: Large-Scale Crowd Intelligence & Parametric Rendering*

The coursework is set within a stylised, abstract interpretation of a Durham landmark (e.g., *The Botanic Garden* or *Riverbanks*) transformed for a light festival. The scene must simulate a massive convergence of "Light Agents" interacting with **procedurally generated scenery composed of dynamic parametric surfaces and reactive geometric forms**.

You are required to develop and implement a high-performance graphics simulation capable of handling **1,000+ active entities** at real-time framerates (60 FPS). The assessment focuses on how you research and orchestrate conflicting technical objectives: **Complex Swarm Behaviour vs. Rendering Optimisation**.

This coursework aims for you to demonstrate your learning outcomes through designing and implementing advanced computer graphics (ACG) methods. There are important points to note:

1. Marks are allocated for correct implementations / answers to each question, with partial credit available for good attempts. Note that the assessment of Questions 2 to 4 will be based on the implemented code and the runtime execution of the required ACG application.
2. The level of achievement (good/very good/excellent/etc.) for each marking criterion is determined based on the marking and classification conventions published in the university core regulations (pp 15-16): [link](#).
3. Failure to submit the two-page A4 report and required video for Question 1 will result in a zero mark for the coursework.
4. You must author the core simulation algorithms, procedural generation, and rendering architecture yourself. Using generative AI for these assessed technical elements is prohibited. Reliance on AI for assessed logic will be treated as plagiarism.
5. A FAQ section in Blackboard Ultra will be updated with questions as they arise.
6. Before proceeding, please attend all lectures, formative exercise class, Q&A sessions, conduct Three.js programming tutorials and carefully read all instructions.

#### Question 1: Graphics Application Design (16 Marks)

*Focus: Communication, Critical Reflection & System Architecture.*

To effectively communicate your design and implementation of the required simulation, you are required to:

##### **a) Produce a 1-minute video (6 marks)**

Produce a video walkthrough demonstrating your simulation in action. You must articulate the specific Advanced Computer Graphics (ACG) methods implemented, explaining how they synergise to optimise rendering performance, elevate visual quality, and support real-time interactivity.

##### **b) Write a two-page A4 report (10 marks)**

Write a report that provides a clear explanation for your implementation of Questions 2 to 4, explicitly demonstrating how your simulation fulfils the marking criteria. Instead of merely listing features, structure your critical evaluation around the following technical pillars:

- **Performance & Scalability:** Discuss how your simulation handles increasing numbers of "Light Agents" (from hundreds to thousands). Explain the theoretical performance limitations of a basic "brute-force" approach (checking every agent against every other agent) and justify how your implemented solution improves efficiency to maintain a smooth frame rate (e.g.,  $O(N)$  vs  $O(N^2)$  complexity).
- **Optimisation Strategy:** Justify your choice of acceleration structures (e.g., Quadtrees, or Grids) used in Question 3. Why was your specific choice appropriate for the density and distribution of this specific "Lumiere" scene? Contrast this with other potential structures you rejected.
- **Integration Logic:** Discuss how your simulation loop (CPU) communicates with your rendering pipeline (GPU). Specifically, explain the data flow involved in Hardware Instancing and Parametric Geometry generation. How did you minimise bus traffic (CPU-to-GPU bandwidth) when updating the transforms and colours of thousands of active objects every frame?

**Note:** If you did not address Questions 2 to 4, Question 1 will receive a mark of zero.

## Question 2: Parametric Environments & Spatial Awareness (24 Marks)

*Focus: Geometry & Data Structures.*

Construct the "Stage" for your simulation. This must be a procedural representation of a Durham environment (e.g., undulating garden terrain or geometric cathedral arches).

- Parametric Modelling:** Rather than loading external 3D assets, implement algorithms to generate the environment geometry procedurally. The structure should be mathematically defined (e.g., using noise functions, parametric curves/surfaces) to create organic or architectural forms suitable for agent navigation. (12 marks)
- Spatial Discretisation:** To support the massive scale required in Question 3, you must implement and visualise a spatial partitioning data structure. Include a "Debug View" that renders the underlying grid/tree structure, visually demonstrating how the world space is subdivided for collision and query Optimisation. (12 marks)

## Question 3: High-Density Crowd Simulation & Kinematics (40 Marks)

*Focus: Animation, Optimisation & Instancing.*

Populate the environment with a massive swarm of at least **1,000 autonomous "Light Agents."**

- Crowd Intelligence:** Implement a behavioural model where agents navigate the environment autonomously. They must demonstrate group behaviours (e.g., flocking, queuing, or flow fields) while avoiding collisions with the procedural environment generated in Question 2. (15 marks)
- Procedural Kinematics:** The agents must utilise procedural animation techniques to react to their velocity or state. Implement a system (e.g., Forward Kinematics or Inverse Kinematics) where the agent's geometry transforms dynamically (e.g., banking turns, compression on impact, or reaching towards targets) without relying on pre-baked animation clips. (10 marks)
- Rendering Optimisation Pipeline:** To maintain performance with 1,000+ animated entities, you must implement an efficient rendering pipeline. Utilising hardware-accelerated batching techniques (such as Instanced Rendering) is mandatory. You must demonstrate how you efficiently update the transformation matrices of the swarm on the GPU side. (15 marks)

#### Question 4: Adaptive Rendering & Visual Signal Processing (20 Marks)

*Focus: Aliasing/Signals & Rendering Techniques.*

Enhance the visual fidelity of the "Lumiere" theme while managing the cost of rendering high-density scenes.

**a) Adaptive Level of Detail (LOD):** Implement a strategy to balance visual quality with performance based on camera proximity. Also implement a technique (e.g., Discrete LOD, Impostors, or Adaptive Tessellation) that reduces the geometric complexity of distant agents or environmental features without causing jarring visual "popping." (10 marks)

**b) Signal Processing & Light Transport:** Address the artifacts inherent in digital sampling.

- Demonstrate a solution for **Aliasing** (spatial or temporal) evident in your high-frequency geometry.
- Implement a post-processing or shader-based lighting technique (e.g., Bloom, Screen-Space Reflections, etc.) to unify the "Light Festival" aesthetic. (10 marks)

--- The End ---